1.52 3D Vectors: The Scalar Product

Worked Solutions

1 a)

$$|\mathbf{a}| = \sqrt{3^2 + 5^2 + (-2)^2} = \sqrt{38}.$$

1 b)

$$|\mathbf{b}| = \sqrt{0^2 + 12^2 + (-16)^2} = \sqrt{400} = 20.$$

2 a)

$$\mathbf{i} \cdot \mathbf{j} = 0.$$

2 b)

$$\mathbf{j} \cdot \mathbf{k} = 0.$$

2 c)

$$\mathbf{i} \cdot \mathbf{i} = 1.$$

2 d)

$$5\mathbf{j} \cdot 3\mathbf{j} = 15.$$

2 e)

$$2\mathbf{j} \cdot 6\mathbf{k} = 0.$$

2 f)

$$\mathbf{j} \cdot (\mathbf{i} + \mathbf{j}) = \mathbf{j} \cdot \mathbf{i} + \mathbf{j} \cdot \mathbf{j} = 0 + 1 = 1.$$

2 g)

$$\mathbf{k} \cdot (2\mathbf{i} - \mathbf{j}) = \mathbf{k} \cdot 2\mathbf{i} - \mathbf{k} \cdot \mathbf{j} = 0 - 0 = 0.$$

2 h)

$$\mathbf{i} \cdot (3\mathbf{i} - 2\mathbf{j} + \mathbf{k}) = \mathbf{i} \cdot 3\mathbf{i} - \mathbf{i} \cdot 2\mathbf{j} + \mathbf{i} \cdot \mathbf{k} = 3 - 0 + 0 = 3.$$

2 i)

$$(\mathbf{i}+\mathbf{j})\cdot(\mathbf{i}+\mathbf{k}) = \mathbf{i}\cdot\mathbf{i}+\mathbf{i}\cdot\mathbf{k}+\mathbf{j}\cdot\mathbf{i}+\mathbf{j}\cdot\mathbf{k} = 1+0+0+0 = 1.$$

2 j)

$$(2\mathbf{i}-3\mathbf{k})\cdot(2\mathbf{j}-3\mathbf{k}) = 2\mathbf{i}\cdot 2\mathbf{j}-2\mathbf{i}\cdot 3\mathbf{k}-3\mathbf{k}\cdot 2\mathbf{j}+3\mathbf{k}\cdot 3\mathbf{k} = 0-0+0+9 = 9.$$

2 k)

$$\begin{gathered}(\mathbf{i}-\mathbf{j}-\mathbf{k})\cdot(\mathbf{i}+2\mathbf{j}-3\mathbf{k})\\ = \mathbf{i}\cdot\mathbf{i}+\mathbf{i}\cdot 2\mathbf{j}-\mathbf{i}\cdot 3\mathbf{k}-\mathbf{j}\cdot 2\mathbf{j}+\mathbf{j}\cdot 3\mathbf{k}-\mathbf{k}\cdot\mathbf{i}-\mathbf{k}\cdot 2\mathbf{j}+\mathbf{k}\cdot 3\mathbf{k}\\ = 1+0-0-2+0-0-0+3 = 2.\end{gathered}$$

3 a)

$$\mathbf{a}\cdot\mathbf{b} = 4\times 10+2\times 20+3\times(-5) = 65.$$

3 b)

$$\mathbf{a}\cdot\mathbf{b} = 6\times(-3)+1\times 2+(-1)\times(-5) = -11.$$

3 c)

$$\mathbf{a}\cdot\mathbf{b} = 1\times 5+(-1)\times 2+3\times(-1) = 0.$$

3 d)

$$\mathbf{a}\cdot\mathbf{b} = 8\times 16+0\times(-18)+(-15)\times 0 = 128.$$

4)

$$\begin{gathered}(2\mathbf{i}+p\mathbf{j}-3p\mathbf{k})\cdot(3p\mathbf{i}-2\mathbf{j}-\mathbf{k}) = 2\times 3p+p\times(-2)+(-3p)\times(-1)\\ = 6p-2p+3p = 7p.\end{gathered}$$

5)

$$\begin{aligned}\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} &= \frac{1\times(-1)+1\times 3+0\times 0}{\sqrt{1^2+1^2+0^2}\sqrt{(-1)^2+3^2+0^2}}\\ &= \frac{2}{\sqrt{2}\sqrt{10}} = \frac{\sqrt{5}}{5}.\end{aligned}$$

6 a)

$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{1\times 2+3\times 4+4\times 6}{\sqrt{1^2+3^2+4^2}\sqrt{2^2+4^2+6^2}} = \frac{38}{\sqrt{26}\sqrt{56}} = \frac{19\sqrt{91}}{182}.$$

$$\theta = \arccos\left(\frac{19\sqrt{91}}{182}\right) = 5.2^\circ \text{ (1 DP)}.$$

6 b)

$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{3\times 5+2\times(-1)+(-3)\times 2}{\sqrt{3^2+2^2+(-3)^2}\sqrt{5^2+(-1)^2+2^2}}$$

$$= \frac{7}{\sqrt{22}\sqrt{30}} = \frac{7\sqrt{165}}{330}.$$

$$\theta = \arccos\left(\frac{7\sqrt{165}}{330}\right) = 74.2^\circ \text{ (1 DP)}.$$

6 c)

$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{\sqrt{2}\times(-3)+3\times 5\sqrt{2}+2\sqrt{2}\times 5}{\sqrt{(\sqrt{2})^2+3^2+(2\sqrt{2})^2}\sqrt{(-3)^2+(5\sqrt{2})^2+5^2}}$$

$$= \frac{22\sqrt{2}}{\sqrt{19}\sqrt{84}} = \frac{11\sqrt{798}}{399}.$$

$$\theta = \arccos\left(\frac{11\sqrt{798}}{399}\right) = 38.8^\circ \text{ (1 DP)}.$$

6 d)

$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} = \frac{\frac{1}{\sqrt{2}}\times\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\times(-\frac{1}{\sqrt{2}})+(-1)\times\frac{1}{\sqrt{2}}}{\sqrt{(\frac{1}{\sqrt{2}})^2+(\frac{1}{\sqrt{2}})^2+(-1)^2}\sqrt{(\frac{1}{\sqrt{2}})^2+(-\frac{1}{\sqrt{2}})^2+(\frac{1}{\sqrt{2}})^2}}$$

$$= \frac{-\frac{1}{\sqrt{2}}}{\sqrt{2}\frac{\sqrt{3}}{2}} = -\frac{\sqrt{6}}{6}.$$

$$\theta = \arccos\left(-\frac{\sqrt{6}}{6}\right) = 114.094\ldots^\circ.$$

Acute angle $= 180^\circ - 114.094\ldots^\circ = 65.9^\circ$ (1 DP).

7)

$$\cos\theta = \frac{\mathbf{a}\cdot\mathbf{b}}{|\mathbf{a}||\mathbf{b}|} \Rightarrow \cos\theta = \frac{8}{2\times\frac{11}{2}} = \frac{8}{11}.$$

$$\therefore \sin\theta = \frac{\sqrt{57}}{11}.$$

8)

$$\overrightarrow{OA} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k} \Rightarrow \overrightarrow{AO} = -2\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}.$$

$$\overrightarrow{OB} = \mathbf{i} - \mathbf{j} - \mathbf{k}.$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = -\mathbf{i} + 2\mathbf{j} - 7\mathbf{k}.$$

$$\cos\theta = \frac{\overrightarrow{AO} \cdot \overrightarrow{AB}}{|\overrightarrow{AO}||\overrightarrow{AB}|} = \frac{(-2)(-1) + 3 \times 2 + (-6)(-7)}{\sqrt{(-2)^2 + 3^2 + (-6)^2}\sqrt{(-1)^2 + 2^2 + (-7)^2}} = \frac{50}{\sqrt{49}\sqrt{54}} = \frac{25\sqrt{6}}{63}.$$

$$\theta = \arccos\left(\frac{25\sqrt{6}}{63}\right) = 13.6^\circ \text{ (1 DP)}.$$

9 a)

$$\overrightarrow{OA} = -4\mathbf{i} + \mathbf{j} - 8\mathbf{k}, \qquad \overrightarrow{OB} = 2\mathbf{i} + 4\mathbf{j} - 4\mathbf{k}.$$

$$\cos\theta = \frac{\overrightarrow{OA} \cdot \overrightarrow{OB}}{|\overrightarrow{OA}||\overrightarrow{OB}|} = \frac{(-4) \times 2 + 1 \times 4 + (-8) \times (-4)}{\sqrt{(-4)^2 + 1^2 + (-8)^2}\sqrt{2^2 + 4^2 + (-4)^2}} = \frac{28}{\sqrt{81}\sqrt{36}} = \frac{14}{27}.$$

9 b)

$$\sin\theta = \frac{\sqrt{533}}{27}.$$

$$\text{Area} = \frac{1}{2}|\overrightarrow{OA}||\overrightarrow{OB}|\sin\theta = \frac{1}{2} \times \sqrt{81} \times \sqrt{36} \times \frac{\sqrt{533}}{27} = \sqrt{533} \text{ units}^2.$$

10)

$$\overrightarrow{OA} = -3\mathbf{i} - \mathbf{j} + 2\mathbf{k}, \quad \overrightarrow{OB} = 8\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}, \quad \overrightarrow{OC} = 2\mathbf{i} - 9\mathbf{j} - 10\mathbf{k}.$$

$$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = -11\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}.$$

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = -6\mathbf{i} - 5\mathbf{j} - 14\mathbf{k}.$$

$$\cos\theta = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{|\overrightarrow{BA}||\overrightarrow{BC}|} = \frac{(-11)(-6) + 3(-5) + (-2)(-14)}{\sqrt{(-11)^2 + 3^2 + (-2)^2}\sqrt{(-6)^2 + (-5)^2 + (-14)^2}}$$

$$= \frac{79}{\sqrt{134}\sqrt{257}} = \frac{79\sqrt{34438}}{34438}.$$

$$\theta = \arccos\left(\frac{79\sqrt{34438}}{34438}\right) = 64.8^\circ \text{ (1 DP)}.$$

11 a)

$$\mathbf{a} \cdot \mathbf{b} = 6 \times 4 + (-1) \times 4 + 4 \times (-5) = 0.$$

$\therefore$ $\mathbf{a}$ and $\mathbf{b}$ are perpendicular.

11 b)

$$\mathbf{a} \cdot \mathbf{b} = (-2) \times 3 + 4 \times (-6) + 5 \times (-6) = -60 \neq 0.$$

$\therefore$ **a** and **b** are not perpendicular.

11 c)

$$\mathbf{a} \cdot \mathbf{b} = 9 \times (-5) + (-6) \times (-10) + 15 \times (-1) = 0.$$

$\therefore$ **a** and **b** are perpendicular.

12 a)

$$\overrightarrow{OA} = 8\mathbf{i} - 2\mathbf{j} + \mathbf{k}, \quad \overrightarrow{OB} = 10\mathbf{i} - 6\mathbf{j} + 5\mathbf{k}, \quad \overrightarrow{OC} = 10\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}.$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = 2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = 2\mathbf{i} - 2\mathbf{j} + \mathbf{k}.$$

$$\cos\theta = \frac{\overrightarrow{AB} \cdot \overrightarrow{AC}}{|\overrightarrow{AB}||\overrightarrow{AC}|} = \frac{2 \times 2 + (-4)(-2) + 4 \times 1}{\sqrt{2^2 + (-4)^2 + 4^2}\sqrt{2^2 + (-2)^2 + 1^2}} = \frac{16}{\sqrt{36}\sqrt{9}} = \frac{8}{9}.$$

$$\therefore \sin\theta = \frac{\sqrt{17}}{9}.$$

12 b)

$$\overrightarrow{OD} = 12\mathbf{i} + 4\mathbf{j} + 5\mathbf{k}.$$

$$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA} = 4\mathbf{i} + 6\mathbf{j} + 4\mathbf{k}.$$

$$\overrightarrow{AD} \cdot \overrightarrow{AB} = 4 \times 2 + 6 \times (-4) + 4 \times 4 = 0.$$

$$\overrightarrow{AD} \cdot \overrightarrow{AC} = 4 \times 2 + 6 \times (-2) + 4 \times 1 = 0.$$

$\therefore \overrightarrow{AD}$ is perpendicular to $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

12 c)

$$\text{Area of triangle } ABC = \frac{1}{2}|\overrightarrow{AB}||\overrightarrow{AC}|\sin\theta = \frac{1}{2} \times \sqrt{36} \times \sqrt{9} \times \frac{\sqrt{17}}{9} = \sqrt{17}.$$

$$|\overrightarrow{AD}| = \sqrt{4^2 + 6^2 + 4^2} = \sqrt{68} = 2\sqrt{17}.$$

$$\text{Volume of Pyramid} = \frac{1}{3} \times \sqrt{17} \times 2\sqrt{17} = \frac{34}{3} \text{ units}^3.$$

13 a)

$$\overrightarrow{OA} = \mathbf{i} + 9\mathbf{j} - 6\mathbf{k}, \quad \overrightarrow{OB} = 2\mathbf{i} + 5\mathbf{j} + 2\mathbf{k}, \quad \overrightarrow{OC} = \frac{11}{3}\mathbf{i} + \frac{28}{3}\mathbf{j} + \frac{7}{12}\mathbf{k}.$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \mathbf{i} - 4\mathbf{j} + 8\mathbf{k}.$$

$$\overrightarrow{OD} = \overrightarrow{OA} + \frac{2}{3}\overrightarrow{AB} = (\mathbf{i} + 9\mathbf{j} - 6\mathbf{k}) + \frac{2}{3}(\mathbf{i} - 4\mathbf{j} + 8\mathbf{k}) = \frac{5}{3}\mathbf{i} + \frac{19}{3}\mathbf{j} - \frac{2}{3}\mathbf{k}.$$

$$\overrightarrow{DC} = \overrightarrow{OC} - \overrightarrow{OD} = 2\mathbf{i} + 3\mathbf{j} + \frac{5}{4}\mathbf{k}.$$

$$\overrightarrow{AB} \cdot \overrightarrow{DC} = 1 \times 2 + (-4) \times 3 + 8 \times \frac{5}{4} = 0.$$

$\therefore \overrightarrow{AB}$ and $\overrightarrow{DC}$ are perpendicular.

13 b)

$$|\overrightarrow{AB}| = \sqrt{1^2 + (-4)^2 + 8^2} = \sqrt{81} = 9.$$

$$|\overrightarrow{DC}| = \sqrt{2^2 + 3^2 + (\frac{5}{4})^2} = \sqrt{\frac{233}{16}} = \frac{\sqrt{233}}{4}.$$

$$\text{Area of triangle } ABC = \frac{1}{2}|\overrightarrow{AB}||\overrightarrow{DC}| = \frac{1}{2} \times 9 \times \frac{\sqrt{233}}{4} = \frac{9\sqrt{233}}{8} \text{ units}^2.$$

14 a)

$$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow 3p \times 2 + 2 \times (-4p) + (-1) \times 6 = 0$$

$$\Rightarrow 6p - 8p - 6 = 0 \Rightarrow -2p - 6 = 0 \Rightarrow p = -3.$$

14 b)

$$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow 16 \times (-3) + (-4p) \times 7 + 7 \times 40p = 0$$

$$\Rightarrow -48 - 28p + 280p = 0 \Rightarrow -48 + 252p = 0 \Rightarrow p = \frac{4}{21}.$$

14 c)

$$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow 5p \times p + (-p) \times (-5) + 1 \times 8p = 0$$

$$\Rightarrow 5p^2 + 5p + 8p = 0 \Rightarrow 5p^2 + 13p = 0 \Rightarrow p(5p + 13) = 0 \Rightarrow p = 0 \text{ or } p = -\frac{13}{5}.$$

14 d)

$$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow 4p \times (-3p) + (-6p) \times 2 + 12 \times 6 = 0$$

$$\Rightarrow -12p^2 - 12p + 72 = 0 \Rightarrow p^2 + p - 6 = 0 \Rightarrow (p + 3)(p - 2) = 0 \Rightarrow p = -3 \text{ or } p = 2.$$

15 a)

$$\overrightarrow{OA} = 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}, \quad \overrightarrow{OB} = \mathbf{i} + 5\mathbf{j} + 4\mathbf{k}, \quad \overrightarrow{OC} = 6\mathbf{i} + (p-5)\mathbf{j} - \mathbf{k}.$$

$$\overrightarrow{CA} = \overrightarrow{OA} - \overrightarrow{OC} = -4\mathbf{i} + (4-p)\mathbf{j} + 4\mathbf{k}.$$

$$\overrightarrow{CB} = \overrightarrow{OB} - \overrightarrow{OC} = -5\mathbf{i} + (10-p)\mathbf{j} + 5\mathbf{k}.$$

Angle $ACB = 90° \Rightarrow \overrightarrow{CA} \cdot \overrightarrow{CB} = 0.$

$$\therefore (-4) \times (-5) + (4-p)(10-p) + 4 \times 5 = 0$$

$$\Rightarrow 20 + 40 - 14p + p^2 + 20 = 0 \Rightarrow p^2 - 14p + 80 = 0.$$

$$b^2 - 4ac = (-14)^2 - 4 \times 1 \times 80 = -124 < 0.$$

So angle ACB can never be $90°$.

15 b)

$$\overrightarrow{BA} = \overrightarrow{OA} - \overrightarrow{OB} = \mathbf{i} - 6\mathbf{j} - \mathbf{k}.$$

$$\overrightarrow{BC} = \overrightarrow{OC} - \overrightarrow{OB} = 5\mathbf{i} + (p-10)\mathbf{j} - 5\mathbf{k}.$$

Angle $ABC = 90° \Rightarrow \overrightarrow{BA} \cdot \overrightarrow{BC} = 0.$

$$\therefore 1 \times 5 + (-6)(p-10) + (-1)(-5) = 0$$

$$\Rightarrow 5 - 6p + 60 + 5 = 0 \Rightarrow 70 - 6p \Rightarrow p = \frac{35}{3}.$$

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = -\mathbf{i} + 6\mathbf{j} + \mathbf{k}.$$

$$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = 4\mathbf{i} + (p-4)\mathbf{j} - 4\mathbf{k}.$$

If A, B and C are collinear then $\overrightarrow{AB} = k\overrightarrow{AC}$ for some constant k.

$$-1 = 4k \Rightarrow k = -\frac{1}{4}, \qquad 6 = (p-4)k, \qquad 1 = -4k \Rightarrow k = -\frac{1}{4}.$$

$\therefore 6 = (p-4) \times (-\frac{1}{4}) \Rightarrow p = -20.$

16)

$$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow \begin{pmatrix} 3k \\ -2 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ -3 \\ -2 \end{pmatrix} = 0 \Rightarrow (3k) \times 4 + (-2) \times (-3) + 5 \times (-2) = 0$$

$$\Rightarrow 12k + 6 - 10 = 0 \Rightarrow k = \frac{1}{3}.$$

17)

$$\mathbf{a} \cdot \mathbf{b} = 0 \Rightarrow \begin{pmatrix} -6 \\ 2(k+1)^2 \\ 29 \end{pmatrix} \cdot \begin{pmatrix} 8 \\ -5 \\ 2 \end{pmatrix} = 0$$

$$\Rightarrow (-6) \times 8 + 2(k+1)^2 \times (-5) + 29 \times 2 = 0$$

$$\Rightarrow -48 - 10(k+1)^2 + 58 = 0 \Rightarrow (k+1)^2 = 1 \Rightarrow k + 1 = \pm 1 \Rightarrow k = -2 \text{ or } k = 0.$$

18)

$$\mathbf{a}\cdot\mathbf{c}=0\Rightarrow 2\times 1+3\times m+5\times n=0\Rightarrow 3m+5n=-2\quad(1)$$
$$\mathbf{b}\cdot\mathbf{c}=0\Rightarrow 5\times 1+4\times m+2\times n=0\Rightarrow 4m+2n=-5\quad(2).$$

$\therefore m=-\frac{3}{2}$ and $n=\frac{1}{2}$.

19)

$$\mathbf{a}\cdot\mathbf{c}=0\Rightarrow\begin{pmatrix}-13\\-4\\-9\end{pmatrix}\cdot\begin{pmatrix}m\\n\\5\end{pmatrix}=0\Rightarrow -13m-4n=45\quad(1)$$
$$\mathbf{b}\cdot\mathbf{c}=0\Rightarrow\begin{pmatrix}3\\-7\\10\end{pmatrix}\cdot\begin{pmatrix}m\\n\\5\end{pmatrix}=0\Rightarrow 3m-7n=-50\quad(2).$$

$\therefore m=-5$ and $n=5$.

20)

Let $\mathbf{c}=\mathbf{i}+m\mathbf{j}+n\mathbf{k}$ so that $\mathbf{a}\cdot\mathbf{c}=0$ and $\mathbf{b}\cdot\mathbf{c}=0$.

$$\mathbf{a}\cdot\mathbf{c}=0\Rightarrow 1\times 1+(-2)\times m+3\times n=0\Rightarrow -2m+3n=-1\quad(1)$$

$$\mathbf{b}\cdot\mathbf{c}=0\Rightarrow 4\times 1+(-5)\times m+2\times n=0\Rightarrow -5m+2n=-4\quad(2).$$

$\therefore m=\frac{10}{11}$ and $n=\frac{3}{11}$. So $\mathbf{c}=\mathbf{i}+\frac{10}{11}\mathbf{j}+\frac{3}{11}\mathbf{k}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

21)

Let $\mathbf{c}=\mathbf{i}+m\mathbf{j}+n\mathbf{k}$ so that $\mathbf{a}\cdot\mathbf{c}=0$ and $\mathbf{b}\cdot\mathbf{c}=0$.

$$\mathbf{a}\cdot\mathbf{c}=0\Rightarrow(-42)\times 1+(-1)\times m+6\times n=0\Rightarrow -m+6n=42\quad(1)$$

$$\mathbf{b}\cdot\mathbf{c}=0\Rightarrow 10\times 1+5\times m+(-5)\times n=0\Rightarrow 5m-5n=-10\quad(2).$$

$\therefore m=6$ and $n=8$. So $\mathbf{c}=\mathbf{i}+6\mathbf{j}+8\mathbf{k}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

$$\hat{\mathbf{c}}=\frac{\mathbf{c}}{|\mathbf{c}|}=\frac{1}{\sqrt{1^2+6^2+8^2}}(\mathbf{i}+6\mathbf{j}+8\mathbf{k})=\frac{1}{\sqrt{101}}(\mathbf{i}+6\mathbf{j}+8\mathbf{k})$$
$$=\frac{1}{\sqrt{101}}\mathbf{i}+\frac{6}{\sqrt{101}}\mathbf{j}+\frac{8}{\sqrt{101}}\mathbf{k}$$

is a unit vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

22)

Let $\mathbf{c}=\begin{pmatrix}1\\m\\n\end{pmatrix}$ so that $\mathbf{a}\cdot\mathbf{c}=0$ and $\mathbf{b}\cdot\mathbf{c}=0$.

$$\mathbf{a}\cdot\mathbf{c}=0\Rightarrow p\times 1+3p\times m+5\times n=0\Rightarrow p+3pm+5n=0\quad(1)$$

$$\mathbf{b}\cdot\mathbf{c}=0\Rightarrow(-2)\times 1+4\times m+3\times n=0\Rightarrow -2+4m+3n=0\quad(2).$$

$$(1)\times 3:\quad 3p+9pm+15n=0\quad(3)$$

$$(2) \times 5: \quad -10 + 20m + 15n = 0 \quad (4)$$

$$(3) - (4): \quad 3p + 9pm + 10 - 20m = 0 \Rightarrow 3p + 10 = (20 - 9p)m \Rightarrow m = \frac{3p+10}{20-9p}.$$

$$\therefore -2 + 4 \times \frac{3p+10}{20-9p} + 3n = 0 \Rightarrow n = \frac{1}{3}\left(2 - \frac{12p+40}{20-9p}\right)$$

$$= \frac{1}{3}\left(\frac{40-18p}{20-9p} - \frac{12p+40}{20-9p}\right) = \frac{1}{3}\left(\frac{-30p}{20-9p}\right) = -\frac{10p}{20-9p}.$$

So $\mathbf{c} = \begin{pmatrix} 1 \\ \frac{3p+10}{20-9p} \\ -\frac{10p}{20-9p} \end{pmatrix}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. An alternative is $\begin{pmatrix} 20-9p \\ 3p+10 \\ -10p \end{pmatrix}$.

23)

Let $\mathbf{c} = \begin{pmatrix} 1 \\ m \\ n \end{pmatrix}$ so that $\mathbf{a} \cdot \mathbf{c} = 0$ and $\mathbf{b} \cdot \mathbf{c} = 0$.

$$\mathbf{a} \cdot \mathbf{c} = 0 \Rightarrow 2q \times 1 + (-5) \times m + 3q \times n = 0 \Rightarrow 2q - 5m + 3qn = 0 \quad (1)$$

$$\mathbf{b} \cdot \mathbf{c} = 0 \Rightarrow 5 \times 1 + 7 \times m + 1 \times n = 0 \Rightarrow 5 + 7m + n = 0 \quad (2).$$

$$(1) \times 7: \quad 14q - 35m + 21qn = 0 \quad (3)$$

$$(2) \times 5: \quad 25 + 35m + 5n = 0 \quad (4)$$

$$(3) + (4): \quad 14q + 21qn + 25 + 5n = 0 \Rightarrow (21q+5)n = -(14q+25) \Rightarrow n = -\frac{14q+25}{21q+5}.$$

$$\therefore 5 + 7m - \frac{14q+25}{21q+5} = 0 \Rightarrow m = \frac{1}{7}\left(\frac{14q+25}{21q+5} - 5\right)$$

$$= \frac{1}{7}\left(\frac{14q+25}{21q+5} - \frac{105q+25}{21q+5}\right) = \frac{1}{7}\left(-\frac{91q}{21q+5}\right) = -\frac{13q}{21q+5}.$$

So $\mathbf{c} = \begin{pmatrix} 1 \\ -\frac{13q}{21q+5} \\ -\frac{14q+25}{21q+5} \end{pmatrix}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. An alternative is $\begin{pmatrix} 21q+5 \\ -13q \\ -14q-25 \end{pmatrix}$.

24)

$$\mathbf{a} \cdot \mathbf{b} = \begin{pmatrix} x+1 \\ x-2 \\ 2x+3 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 4x \\ 7 \end{pmatrix}$$

$$= (x+1)(-2) + (x-2)4x + (2x+3)7$$

$$= -2x - 2 + 4x^2 - 8x + 14x + 21 = 4x^2 + 4x + 19 = 4(x^2 + x) + 19$$

$$= 4((x + \frac{1}{2})^2 - \frac{1}{4}) + 19 = 4(x + \frac{1}{2})^2 - 1 + 19 = 4(x + \frac{1}{2})^2 + 18 > 0$$

for all real values of x, so $\mathbf{a} \cdot \mathbf{b} \neq 0$ and so $\mathbf{a}$ and $\mathbf{b}$ can never be perpendicular.

25)

$$\mathbf{a}\cdot\mathbf{b}=\begin{pmatrix}3-x\\4-3x\\5-x\end{pmatrix}\cdot\begin{pmatrix}x-1\\x\\-5\end{pmatrix}$$

$$=(3-x)(x-1)+(4-3x)x+(5-x)(-5)$$

$$=3x-3-x^2+x+4x-3x^2-25+5x=-4x^2+13x-28$$

$$=-4(x^2-\frac{13}{4}x)-28=-4((x-\frac{13}{8})^2-\frac{169}{64})-28$$

$$=-4(x-\frac{13}{8})^2+\frac{169}{16}-28=-4(x-\frac{13}{8})^2-\frac{279}{16}<0$$

for all real values of x, so $\mathbf{a}\cdot\mathbf{b}\neq 0$ and so $\mathbf{a}$ and $\mathbf{b}$ can never be perpendicular.

26 a)

Let $\mathbf{c}=\begin{pmatrix}1\\m\\n\end{pmatrix}$ so that $\mathbf{a}\cdot\mathbf{c}=0$ and $\mathbf{b}\cdot\mathbf{c}=0$.

$$\mathbf{a}\cdot\mathbf{c}=0\Rightarrow 2\times 1+q\times m+(-3)\times n=0\Rightarrow 2+mq-3n=0\quad(1)$$

$$\mathbf{b}\cdot\mathbf{c}=0\Rightarrow (q+1)\times 1+q\times m+(-1)\times n=0\Rightarrow q+1+mq-n=0\quad(2)$$

$$(2)-(1):q-1+2n=0\Rightarrow n=\frac{1}{2}(1-q).$$

$$\therefore q+1+mq-\frac{1}{2}(1-q)=0\Rightarrow q+1+mq-\frac{1}{2}+\frac{1}{2}q=0\Rightarrow\frac{3}{2}q+\frac{1}{2}+mq=0$$

$$\Rightarrow m=-\frac{1}{q}\left(\frac{3}{2}q+\frac{1}{2}\right)\Rightarrow m=-\frac{3q+1}{2q}.$$

So $\mathbf{c}=\begin{pmatrix}1\\-\frac{3q+1}{2q}\\\frac{1}{2}(1-q)\end{pmatrix}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

An alternative is $\begin{pmatrix}2q\\-3q-1\\q-q^2\end{pmatrix}$.

26 b)

$$\mathbf{c}\cdot\mathbf{d}=2q\times(-5q)+(-3q-1)\times 5+(q-q^2)\times 1=-10q^2-15q-5+q-q^2$$

$$=-11q^2-14q-5=-11\left(q^2+\frac{14}{11}q\right)-5=-11\left((q+\frac{7}{11})^2-\frac{49}{121}\right)-5$$

$$=-11(q+\frac{7}{11})^2+\frac{49}{11}-5=-11(q+\frac{7}{11})^2-\frac{6}{11}<0\text{ for all real values of }q,\text{ so }\mathbf{c}\cdot\mathbf{d}\neq 0$$

and so $\mathbf{c}$ and $\mathbf{d}$ can never be perpendicular.

27)

Let $\mathbf{c} = \begin{pmatrix} m \\ n \\ 1 \end{pmatrix}$ so that $\mathbf{a} \cdot \mathbf{c} = 0$ and $\mathbf{b} \cdot \mathbf{c} = 0$.

$$\mathbf{a} \cdot \mathbf{c} = 0 \Rightarrow 2 \times m + (-3) \times n + x \times 1 = 0 \Rightarrow 2m - 3n + x = 0 \quad (1)$$

$$\mathbf{b} \cdot \mathbf{c} = 0 \Rightarrow (-4) \times m + 3 \times n + (x+5) \times 1 = 0 \Rightarrow -4m + 3n + x + 5 = 0 \quad (2)$$

$$(1) + (2) : -2m + 2x + 5 = 0 \Rightarrow m = \frac{1}{2}(2x+5).$$

$$\therefore 2 \times \frac{1}{2}(2x+5) - 3n + x = 0 \Rightarrow n = \frac{1}{3}(2 \times \frac{1}{2}(2x+5) + x) = \frac{1}{3}(3x+5).$$

So $\mathbf{c} = \begin{pmatrix} \frac{1}{2}(2x+5) \\ \frac{1}{3}(3x+5) \\ 1 \end{pmatrix}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

An alternative is $\begin{pmatrix} 6x+15 \\ 6x+10 \\ 6 \end{pmatrix}$.

To find the shortest possible vector, we need to minimise

$$(6x+15)^2 + (6x+10)^2 + 6^2 = 36x^2 + 180x + 225 + 36x^2 + 120x + 100 + 36$$

$$= 72x^2 + 300x + 361 = 72(x^2 + \frac{25}{6}x) + 361 = 72((x + \frac{25}{12})^2 - \frac{625}{144}) + 361$$

$$= 72(x + \frac{25}{12})^2 - \frac{625}{2} + 361 = 72(x + \frac{25}{12})^2 + \frac{97}{2}.$$

This is minimised when $x = -\frac{25}{12}$.

So $\mathbf{c} = \begin{pmatrix} \frac{5}{12} \\ -\frac{5}{12} \\ 1 \end{pmatrix}$ is the shortest possible vector of the required form that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

28)

Let $\mathbf{c} = \begin{pmatrix} 1 \\ m \\ n \end{pmatrix}$ so that $\mathbf{a} \cdot \mathbf{c} = 0$ and $\mathbf{b} \cdot \mathbf{c} = 0$.

$$\mathbf{a} \cdot \mathbf{c} = 0 \Rightarrow 1 \times 1 + x \times m + 4 \times n = 0 \Rightarrow 1 + mx + 4n = 0 \quad (1)$$

$$\mathbf{b} \cdot \mathbf{c} = 0 \Rightarrow 5 \times 1 + (1-x) \times m + 8 \times n = 0 \Rightarrow 5 + m - mx + 8n = 0 \quad (2)$$

$$(1) \times 2 : 2 + 2mx + 8n = 0 \quad (3)$$

$$(3) - (2) : -3 + 3mx - m = 0 \Rightarrow (3x-1)m = 3 \Rightarrow m = \frac{3}{3x-1}.$$

$$\therefore 1 + \frac{3}{3x-1} \times x + 4n = 0 \Rightarrow n = \frac{1}{4}(-\frac{3x}{3x-1} - 1) = -\frac{1}{4}(\frac{3x}{3x-1} + 1) = -\frac{1}{4}(\frac{3x}{3x-1} + \frac{3x-1}{3x-1})$$

$$= -\frac{1}{4}(\frac{6x-1}{3x-1}).$$

So $\mathbf{c} = \begin{pmatrix} 1 \\ \frac{3}{3x-1} \\ -\frac{1}{4}(\frac{6x-1}{3x-1}) \end{pmatrix}$ is an example of a non-zero vector that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

An alternative is $\begin{pmatrix} 3x-1 \\ 3 \\ -\frac{1}{4}(6x-1) \end{pmatrix}$ or $\begin{pmatrix} 4-12x \\ -12 \\ 6x-1 \end{pmatrix}$.

To find the shortest possible vector, we need to minimise

$$(4-12x)^2+(-12)^2+(6x-1)^2 = 144x^2-96x+16+144+36x^2-12x+1$$

$$= 180x^2-108x+161 = 180(x^2-\frac{3}{5}x)+161 = 180((x-\frac{3}{10})^2-\frac{9}{100})+161$$

$$= 180(x-\frac{3}{10})^2-\frac{81}{5}+161 = 180(x-\frac{3}{10})^2+\frac{724}{5}.$$

This is minimised when $x=\frac{3}{10}$.

So $\mathbf{c}=\begin{pmatrix} 1 \\ -30 \\ 2 \end{pmatrix}$ is the shortest possible vector of the required form that is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$.

29)

Let $\mathbf{p}=\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ where not all of a,b,c are zero, and $\mathbf{q}=\begin{pmatrix} d \\ e \\ f \end{pmatrix}$, where not all of d,e,f are zero.

$$\mathbf{p}+\mathbf{q}=\begin{pmatrix} a+d \\ b+e \\ c+f \end{pmatrix}$$

$$\mathbf{p}-\mathbf{q}=\begin{pmatrix} a-d \\ b-e \\ c-f \end{pmatrix}$$

If $\mathbf{p}+\mathbf{q}$ is perpendicular to $\mathbf{p}-\mathbf{q}$ then $(\mathbf{p}+\mathbf{q})\cdot(\mathbf{p}-\mathbf{q})=0$.

$$\begin{pmatrix} a+d \\ b+e \\ c+f \end{pmatrix}\cdot\begin{pmatrix} a-d \\ b-e \\ c-f \end{pmatrix}=0$$

$$\Rightarrow (a+d)(a-d)+(b+e)(b-e)+(c+f)(c-f)=0$$

$$\Rightarrow a^2-d^2+b^2-e^2+c^2-f^2=0$$

$$\Rightarrow a^2+b^2+c^2=d^2+e^2+f^2$$

$$\Rightarrow \sqrt{a^2+b^2+c^2}=\sqrt{d^2+e^2+f^2}$$

$$\Rightarrow |\mathbf{p}|=|\mathbf{q}|.$$